

MSME IDEA HACKATHON

MSME Host Institute- Vishnu Dental College

PestAware

"Smart Pest Detection. Anywhere. Anytime."

PROBLEM STATEMENT

Pest infestations devastate Indian agriculture each year, wiping out ₹50,000 crore (\$6 billion) in crops and cutting yields by 30%. With over 80% of small farmers lacking access to timely pest alerts—and only 18% of rural farms reliably online—millions are left defenseless against preventable losses.

OUR SOLUTION

Our Offline AI-powered system detects pests directly in the field and instantly alerts the farmer with pest type and count. Broader insights based on combined data from nearby farms, and stores all data for future reference and seasonal analysis. Advanced analytics and personalized recommendations help optimize pest control and overall farm management

Target Audience

- Smallholder farmers
- Large farm owners
- Agri-businesses
- Agricultural research institutions
- Government agricultural departments
- Agri co-operatives and FPOs
- NGOs focused on sustainable farming
- CSR-backed rural development programs

FEATURES

Instant Detection

- Real-Time Alerts
- Automated Counting

Affordable Innovation

- Low-Cost Hardware
- Scalable Design

Seamless Connectivity

- Offline Operation
- Local Notifications via app

LoRaWAN Summary:

- Combined insights from multiple nearby farms

User-Centric

- Easy App
- Customizable Settings
- Multi Language Setting

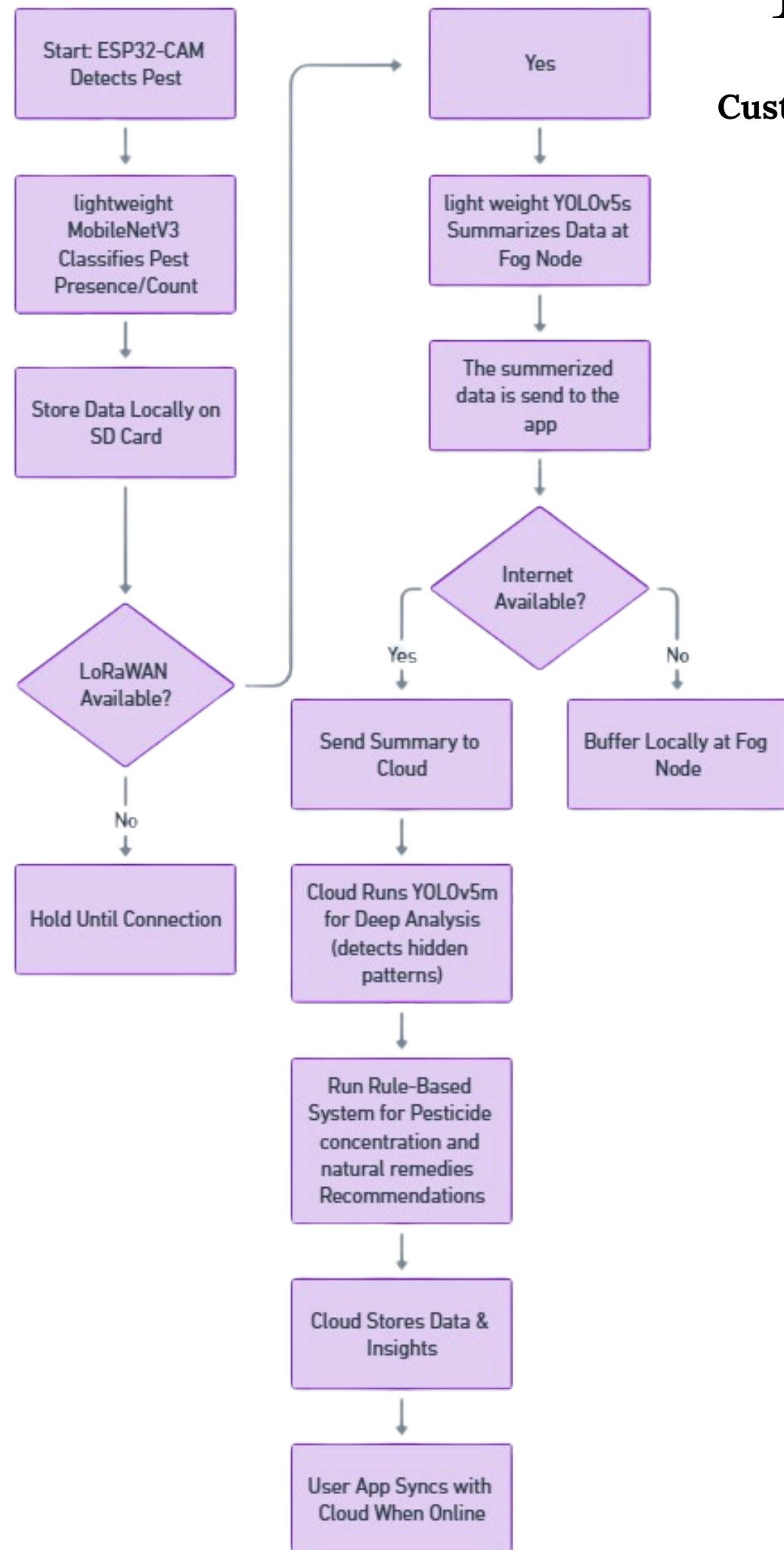
Cloud Intelligence

- Secure Data Backup
- Advanced Analytics

Smart Pest Advice

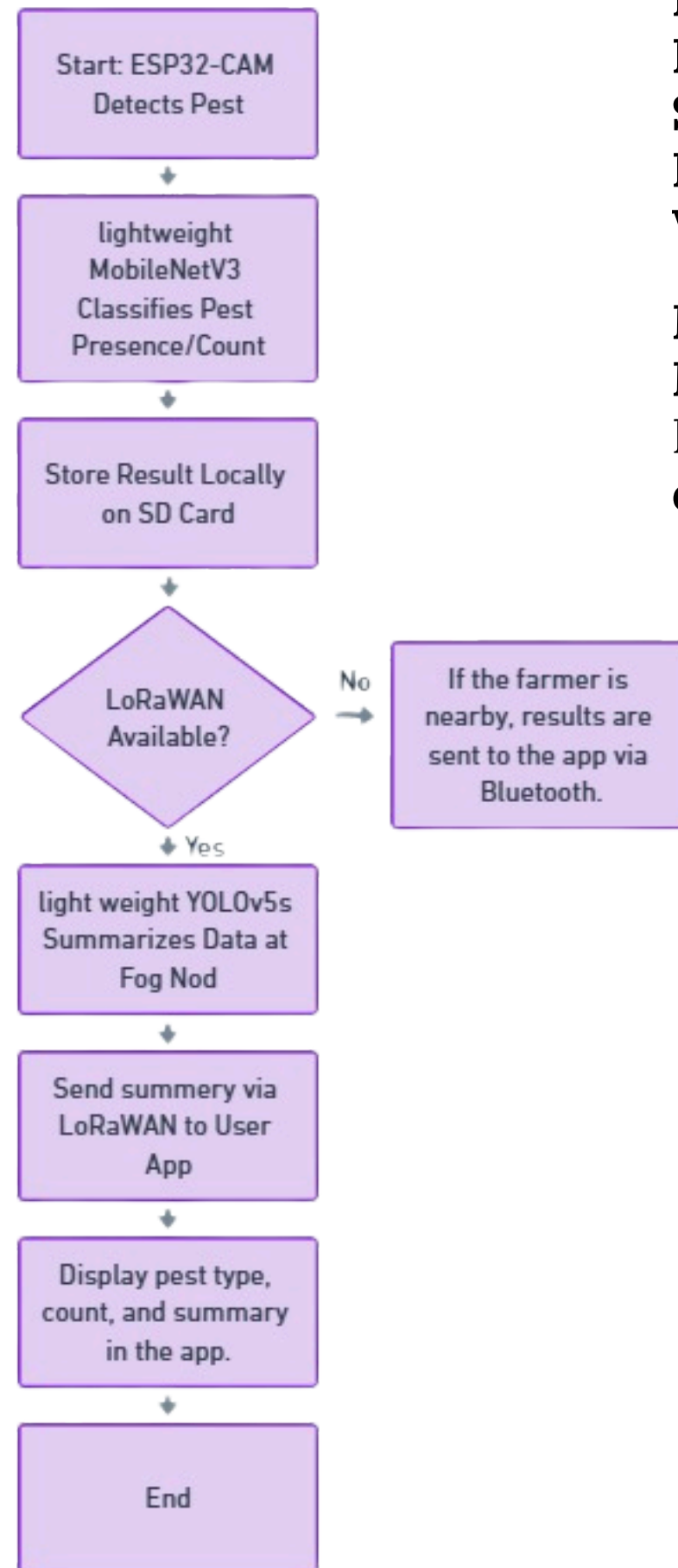
- Pesticide Dose Recommendation
- Natural Remedy Suggestion

Detailed Farm Management Process



TECHNICAL APPROACH

Customizable Process Overview(small Scale Farmers)



ESP32-CAM: Image capture and on-device ML inference.

Battery Pack: Li-ion 18650 or similar, for portable power.

SD Card (8GB): Local storage for images and logs.

LoRa Module: Long-range, low-power communication to fog node.

Weatherproof Enclosure: Field-ready protection.

MobileNetV3 (quantized, lightweight, TensorFlow Lite for Microcontrollers):

Performs pest presence/absence detection and basic counting directly on the ESP32-CAM

Fog Node

Raspberry Pi Zero 2 W: Aggregates data from multiple edge devices, runs more advanced ML.

YOLOv5s (lightweight): Performs multi-pest detection and summary analytics at the cluster/fog level

Cloud Server: Centralized data storage, backup, and advanced analytics.

YOLOv5m\1 (full model): Advanced pest detection, pattern analysis, and historical trend discovery.

Rule-Based System: Provides expert-verified pesticide and remedy recommendations tailored to pest type, severity, and farm context..

Mobile Application: Flutter,
SQLite :Offline-first data storage and sync for personalized user experience

FEASIBILITY AND VIABILITY



Revenue Model

Component	Quantity	Unit Cost (₹)	Total Cost (₹)	Description
Edge Device	1	1,539	1,539	ESP32-CAM, battery, SD card, LoRa module, enclosure
Fog Node	1	1,550 – 2,200	1,550 – 2,200	Raspberry Pi Zero 2 W + 16GB microSD card
Cluster Total	5 Edge Devices + 1 Fog Node	—	9,245 – 9,895	Total cost for a cluster setup

- Annual subscription of ₹2000 for cloud services.
- Research institutions and aligned organizations can access aggregated farm pest data through a subscription model, with farmer permissions, enabling collaborative insights and supporting platform sustainability.

Unique Selling Proposition

Offline AI-powered detection enables farmers to identify pests without internet access, ensuring reliability in remote areas

Community-driven insights and analytics aggregate local farm data for broader pest trend analysis, empowering collective action and early warning

Customizable and low-cost solution makes advanced pest management accessible and adaptable for diverse farm needs, unlike costly, one-size-fits-all competitors

precise pesticide concentration and natural remedy recommendations tailored to diverse farm needs

TOTAL BUDGET

Category	Description	Allocated Budget (₹)
1. Hardware Initial Batch	ESP32-CAM kits + battery + SD cards (5-10 units), Raspberry Pi fog nodes, LoRaWAN gateway, sensors, enclosures, wiring, misc prototyping components	3,50,000
2. Software Development	Minimal embedded firmware, basic ML model integration, simple mobile app prototype (offline notifications, SQLite), limited cloud backend functionality	70,000
3. Cloud Infrastructure	Basic cloud storage, low-scale compute for backup and minimal analytics, push notifications (annual costs)	1,00,000
4. Patents & Registration	Provisional patent filing, trademark, company registration, and legal fees	1,00,000
5. Marketing & Outreach	Essential digital marketing, demo events, promotional materials to reach initial customers	1,20,000
6. Miscellaneous & Admin	Transport, internet, office supplies, contingency for unforeseen expenses	1,00,000
7. Team & Consulting	Part-time ML experts, advisors, consultants for focused development tasks, troubleshooting	1,50,000
8. Scaling & Piloting	Additional hardware units post initial batch, iterative improvements, pilot field testing	1,10,000

| Total Budget | | 10,00,000 |